



CHEMISTRY Department
SIMHADRI

Note No :1



Ref:SMTTP/Chemistry/2025-26/ofc note/460714

Date:

25/10/2025(18:06:19)

Subject:LMI ON COOLING WATER TREATMENT MONITORING AND
CHEMICAL CLEANING OF CONDENSER TUBES

LOCATION MANAGEMENT INSTRUCTION ON - COOLING WATER TREATMENT MONITORING AND CHEMICAL CLEANING OF CONDENSER TUBES:

Revision of LMI LMI/OGN/OPS/CHEM/019 COOLING WATER TREATMENT MONITORING
AND CHEMICAL CLEANING OF CONDENSER TUBES:

The subject LMI is reviewed and made below mentioned changes as per latest OGN.

Following changes have been done from previous version as described below:

- | | | |
|----|---|---|
| 1. | Section- 8.3: Inhibitor Selection
Criteria for cleaning process. | Modified w.r.t. choice of inhibitors. |
| 2. | Section- 10.0: Inhibitor Efficiency | Inhibitor efficiency modified to $\geq 99\%$ for
all metallurgy. |
| 3. | Section- 11.1.3: Acid Injection | Corrosion inhibitor concentration modified. |

Encl:

1. LMI in word and .PDF Formats.

2. Latest **COS-ISO-00-OGN/OPS/CHEM/019 Rev. No.2, July 2025**

Submitted for approval please

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Note No:2

Draft LMI is submitted for comments from concerned departments and for approval by CA please.

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LMI is in line with the OGN. Forwarded for further review.

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LMI reviewed and is in line with the OGN. Forwarded for further review and approval.

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NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

**COOLING WATER TREATMENT MONITORING AND
CHEMICAL CLEANING OF CONDENSER TUBES**

**LMI/OGN/OPS/CHEM/019 Rev. No.: 02
Dated: Oct-2025**

Approved for Implementation by: HOP (SIMHADRI)

LMI APPROVAL DOCUMENT

1	LMI NAME	COOLING WATER TREATMENT MONITORING AND CHEMICAL CLEANING OF CONDENSER TUBES
2	LMI No.	LMI/OGN/OPS/CHEM/019 Rev. No.: 02
3	Reference document	COS-ISO-00-OGN/OPS/CHEM/19 Rev. No.:2, July. 2025
4	SupersededDocument	LMI/COS-ISO-00-GN/OPS/CHEM/019 Revision: 01 June-23
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TITLE	COOLING WATER TREATMENT MONITORING AND CHEMICAL CLEANING OF CONDENSER TUBES
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1.0 INTRODUCTION

The cooling water system has a significant impact on the cycle efficiency because of vacuum produced at the turbine outlet when steam condenses. The cooling water quality varies from source to source and also seasonal variations are observed. Corrosion, scale deposition and fouling invariably pose problems in cooling water system, leading to failures, unscheduled shut-downs and loss in generation.

In order to overcome any adverse effect or liability to the condenser, it is usually necessary to carry out treatment of the water before it is used for cooling in the plant. Cooling water chemistry control is governed by the chemical composition of the available cooling water sources that range from low-conductivity, low-hardness freshwater, to high-hardness freshwater, to sea and brackish water, and cooling tower water. At NTPC Simhadri, in both stages sea water is being used as the cooling water of condenser.

The sea water cooling system can have various tendencies to corrode condenser materials, organic fouling, and mineral scaling. The goal of cooling water treatment is to avoid formation of mineral scale, organic growth, and corrosion. Being a sea water cooling system, the COC of the system is kept very low. There is no separate cooling water treatment regime given for sea water cooled condenser.

2.0 SUPERSEDED DOCUMENTS:

Cooling water treatment monitoring and chemical cleaning of condenser tubes
LMI/OGN/OPS/CHEM019 Revision: 01, Revision Date 05.06.2023

3.0 RECOROS OF REVISION:

1.	Section- 8.3: Inhibitor Selection Criteria for cleaning process.	Modified w.r.t. choice of inhibitors.
2.	Section- 10.0: Inhibitor Efficiency	Inhibitor efficiency modified to $\geq 99\%$ for all metallurgy.
3.	Section- 11.1.3: Acid Injection	Corrosion inhibitor concentration modified.

4.0 SCOPE:

The LMI prescribes the specific instructions to be taken by Operation, maintenance and Chemistry during Cooling water treatment & Monitoring and during condenser chemical cleaning of both the stages of NTPC Simhadri.

This document specifies the methods and procedures required for cooling water treatment and it's monitoring and preservation of CW system and condenser cleaning.

5.0 COOLING WATER TREATMENT AND MONITORING:

Cooling Water System is one of the most important systems in any power plant as it dissipates heat in the most efficient and cheapest way and also conserves the most precious natural resource- water.

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In the context of a power plant's open cooling water system and also under the present era of water conservation (operating CW systems at high COC to optimize specific water consumption), monitoring of chemistry parameters involve the controlled measurement of adverse effects of corrosion, microbiological accumulation, mineral scale, or suspended solids accumulation. In most cases, the chemistry of the make- up water to an open cooling water system as well as the cooling water within the system varies with time, seasonally, day-to-day, and so trending over the long-term period is required. A consistent monitoring program can track this variability and be used to determine treatment needs. In addition, monitoring the concentration of CW treatment chemicals within the system is a necessity to ensure that treatment levels are adequate for reliable and trouble-free operation.

At NTPC Simhadri, the condenser cooling is done by sea water. Though there is cooling tower but due to maintaining very low COC no specific cooling water treatment is recommended by NETRA. Therefore, many of the chemical dosing are not applicable for NTPC Simhadri.

5.1 COOLING WATER TREATMENT CHEMICALS:

S. No	Type of control	Type of chemical	Name of the chemical	Preferred Dosage Rate*	Remarks
1	Microbiological fouling control	Oxidizing biocides	Chlorine	0.5-1.0 ppm	Effective below pH of 8.0
			Sodium Hypochlorite	0.5 – 1.0 ppm	Shock dosing purpose up to 0.8 ppm.

Cooling Water Treatment chemical dosage regime is based on CC-NETRA recommendations. CW treatment regime will be reviewed by CC-NETRA every two years to cater to any sudden changes in station intake water quality thereby affecting CW make-up water.

CW system parameter healthiness as well as maintenance of requisite residual chemicals (Based on CWT regime) will be reviewed by COS.

Dosage rates mentioned are indicative ones, actual requirement may vary based on intake water and cooling water quality along with seasonal variations as recommended by CC-NETRA in CW treatment programme. Dosing rate is to be optimized based on requirement.

As COC is maintained at 1.3 which is very low, no specific Cooling water treatment is recommended by NETRA. Only Liq. Chlorine through tonner or through electro chlorination and sodium Hypochlorite is dosed for controlling the microbiological fouling control.

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5.2 Cooling water system Parameter Monitoring:

S.No	Parameter	UOM	Frequency	Remarks
Common Data				
1	Unit Load	MW	Once/shift	
2	Cooling Water Flow	M3/Hr	Once/shift	
3	Average blow-down	M3/Hr	Once/shift	
Condenser Data				
4	Condenser Vacuum	Kg/cm2	Once/shift	
5	Condenser I/L temperature	Deg. C	Once/shift	
6	Condenser O/L temperature	Deg. C	Once/shift	
7	ΔT	Deg. C	Once/shift	
8	Condenser ΔP	mmwc	Once/shift	
Chemical Parameters				
9	pH at 25 ° C*		Once/day	Make-up & CW
10	Specific Conductivity*	us/cm	Once/day	Make-up & CW
11	Turbidity*#	ntu	Once/day	Make-up & CW
12	FRC	ppm	Once/day	Make-up & CW
13	TDS	ppm	Once/day	Make-up & CW
14	Chloride as Cl ⁻	ppm	Once/day	Make-up & CW
15	Sulphate,as CaCO ₃		Once/day	Make-up & CW
16	Ca-Hardness as CaCO ₃		Once/day	Make-up & CW
17	Mg-Hardness as CaCO ₃		Once/day	Make-up & CW
18	Total Hardness as CaCO ₃		Once/day	Make-up & CW
19	p-Alkalinity as CaCO ₃		Once/day	Make-up & CW
20	m-Alkalinity as CaCO ₃		Once/day	Make-up & CW
21	Silica, as SiO ₂		Once/day	Make-up & CW
22	Iron, as Fe		Once/day	Only CW
23	Ca-COC		Once/day	Only CW
24	Mg-COC		Once/day	Only CW
25	COC Ratio		Once/day	Only CW
26	TSS#	ppm	Once/week	Make-up & CW
27	Organic Matter	ppm	„	„
28	Oil & Grease	ppm	„	Only CW

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5.2.1 MICROBIOLOGICAL ANALYSIS (For Make-up and CW samples):

Table-I:

S.N.	PARAMETERS TO BE TESTED	Unit	Frequency of Testing
1.	Appearance	----	Fortnightly
2.	Total Viable Count	Cfu/ml	Quarterly
3.	Total Bacterial Count	Cfu/ml	Fortnightly
4.	Total Fungal Count	Cfu/ml	Quarterly
5.	Sulphate Reducing Bacteria	Organism/100 ml	Fortnightly
6.	Iron Oxidizing Bacteria	Organism/100 ml	Quarterly

Note: All the above parameters to be tested as per frequency mentioned. In case of any issues with biofouling in the CW system or inadequate biocide dosing, all parameters mentioned in Table-I and Table-II are to be tested for microbiological analysis from any external NABL accredited/Govt laboratory.

Table-II:

S.N	PARAMETERS TO BE TESTED	Unit
1.	Organic Acid Producing Bacteria	Organism/100 ml
2.	Slime Formers: Pseudomonas, Enterobacter	Organism/100 ml
3.	Gas Producing Bacteria: CO ₂ producers, Ammonia producers	Organism/100 ml
4.	Sulphur Oxidising Bacteria: Thiobacillus Thiooxidans, Thiobacillus Thioparus	Organism/100 ml
5.	Nitrifying Bacteria	Organism/100 ml
6.	Total Active Biomass	Cfu/ml

6.0 General Recommendations:

Should maintain all safety requirements for Cl₂/ClO₂ dosing and handling (as per OGN-OPS-CHEM-018). Safety instructions and relevant MSDS (bilingual/trilingual) shall be under display at key locations.

Physical inspection of CT for algae or other fouling.

7.0 Preservation of CW system in case of USD:

CW water sampling to be done weekly for chemical parameters monitoring.

In case of sea water, Condenser water box is to be drained to avoid scaling. This activity is to be performed after stopping barring gear.

NOTES:

Please refer Operation OGN on “Guidelines for Units Under Reserve Shut-down “(Ref. No. OGN- OPS-SYST-038) for further details.

Joint Protocol (By Operation, Chemistry, MMD-OFS, EEMG) is to be maintained for preservation of CW System in case of USD.

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8.0 Chemical cleaning of Condenser Tubes:

Introduction:

The accumulation of unwanted deposits on equipment surfaces is a phenomenon that occurs in virtually all processes in which untreated water is heated. The deposits commonly encountered may be categorized into the following five groups: (a) mineral scales (e.g. CaCO_3 , $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$, $\text{Ca}_3(\text{PO}_4)_2$, Silica scales), (b) suspended matter (e.g. mud or silt), (c) corrosion products (i.e., Fe_2O_3 , Fe_3O_4 , ZnO , CuO), (d) microbiological, and (e) metal-inhibitor salts. The deposition of these materials, especially on heat exchanger surfaces in cooling, systems can cause a number of operational problems such as plugging of pipes and pumps, poor performance of condenser tubes, increased operation costs, lost production due to system downtime, and ultimately heat exchanger failure. Greater water conservation has been a driver for operating industrial water systems at higher cycles of concentrations, thereby increasing the potential for deposit build up on heat exchanger surfaces. Operating industrial water systems under stressed conditions demands a better understanding of system (feed and recirculating) water chemistry as well as the development of new and innovative agents for controlling scale/deposit, corrosion, and biofouling.

Calcium carbonate is by far the most common scale found in cooling water systems which normally results from the breakdown of calcium bicarbonate present in water, and its formation depends upon the total dissolved salts, pH and temperature of the water.

Periodic post-operational acid cleaning is a well-established method to restore the efficiency of condenser lost due to scale deposition. Condenser tube deposit/scale layer increases gradually in operation of condenser under conditions of high alkalinity, high hardness, higher temperatures, low flow, physical fouling, biofouling, etc. Due to scale in condenser tube heat transfer gets affected and the efficiency of condenser decreases. To remove the condenser scale there are different methods but Chemical cleaning is most effective in combination with high pressure water jet cleaning. If the nature of the deposits (mainly Ca/Mg scales) is such that it cannot be removed by mechanical means, we need to go for acid cleaning to remove the formed scales. Chemical cleaning is thus done in order to restore Heat Transfer efficiency and to ensure cleanliness of the condensers.

8.1 Following is a general procedure recommended for carrying out the Chemical Cleaning of Condensers:

Objective:

The objective of Chemical Cleaning is to remove scale deposited on condenser tubes and to restore thermal efficiency of heat exchangers.

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Criteria for Cleaning:

The main criteria for cleaning are an unacceptable fall in condenser efficiency due to deterioration of condenser vacuum or deposition of scale as confirmed by visual inspection, CW parameter analysis and detailed deposit analysis. Generally, the chemical cleaning would be done under unit shut-down condition. HP jet/bullet cleaning are used to remove loose deposits and removal of hard scale require chemical cleaning. It is mandatory to carry out to HP jet cleaning and Flood test after Chemical Cleaning of Condenser.

(Condenser vacuum loss could be either due to dirty tubes because of deposit in condenser tubes or air ingress issue. If the vacuum loss is more than 10 mm of Hg, Site EEMG in consultation with Chemistry should decide and attribute the loss based on CW chemistry parameters and other site- specific operational issues).

8.2 Acid Selection Criteria for cleaning processes:

In most of the cases, condenser cleaning is carried out by Inhibited Sulphamic acid and Hydrochloric acid through "Off-load" method to ensure proper cleaning and safety of the system depending on the nature of hard deposits and condenser tube metallurgy. Selection of acid depends on the type of metallurgy involved, constituents of scale deposits in the condenser, safety considerations, availability of chemicals etc.

Metallurgy of Condenser in Simhadri	Acid used as a Descaling
Titanium	Inhibited Sulphamic Acid

8.3 Inhibitor Selection Criteria for cleaning process

The methods of chemical cleaning impose quite different requirements for inhibition and these are influenced by the material of construction of the plant and by the estimated long-term frequency of cleaning. By the use of inhibitors, acceptable conditions can be achieved in respect of the corrosion of water-boxes and condenser tubes, as well as restraint of galvanic attack and copper deposition and dezincification. Choice of inhibitors shall be based on the descalant type used.

8.4 Cleaning methodology and requisites for chemical cleaning:

An illustrative line diagram of condenser acid cleaning for top-to-top, bottom-to-top and top-to-bottom circulation is given in Annexure-3.

8.4.1 Acid Solution Tank

This envisages installation of a dissolving tank of about 20-25 M³ capacity, fabricated with mild steel plates and provided with overflow, delivery drain and perforated pipe for steam admission. The tank shall be located nearer to the condenser for ease of convenience.

8.4.2 Circulating Pumps

At least two nos. of pumps (with stand by option), each of preferable capacity 200 m³/hr and

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head 20 MWC are to be installed near the acid mixing tank. The starters for the pumps can be installed at suitable distance to prevent any damage in case of flash- over from the pumps during operation. [As large amounts of gas- producing scale (e.g. calcium carbonate), are normally present, high velocities (>0.9 m/s, 3 ft/s) to the condenser tubes are needed to provide a satisfactory liquid cleaning].

8.4.3 Connecting Points

The connecting points are to be as shown in the diagram.

For effective cleaning, top-to-top, bottom-to-top and top-to-bottom circulation provision to be kept. Depending on the quantity of scales formed, cleaning methodology is to be developed. Pressure gauges, temperature indicators, and sample points are to be installed at appropriate places.

Vent provision of dimension as mentioned in the circuit diagram is to be kept on both sides of each half of the condenser as shown. Vent return points to be positioned in such a way that it is visible from ground points.

Drain of the system is to be extended to the neutralization pit for neutralising condenser cleaning effluents before final disposal.

8.4.4 Other Arrangements required for Chemical Cleaning

Following activities need to be completed before commencement of chemical cleaning operation of condensers.

Supply of water to the acid solution tank and DM water condenser hot well

Temporary line is to be tapped off from DM water line and is to be led to the acid solution tank with a valve. The make-up flow rate to the acid tank should be preferably 80-100 M³/hr. Arrangements for filling the condenser (condensing side) with DM water are also to be made to prevent any damage to the condenser in case of any leakage. Clarified water is also required for high pressure water jet cleaning. So, the arrangement for water points for all the above purposes is required to be made accordingly.

8.4.5 Supply of steam to acid Solution Tank:

The required quantity of steam for heating the acid solution is to be supplied preferably through 1.5-2inch seamless pipe or more as per requirement of cleaning. The other parameters shall be preferably as follows.

Pressure – 8 to 10 kg/cm² Temperature – 150 to 200° C

Two tapping points from the existing steam source shall be taken (for quick heating). The steam line is to be led into the acid solution tank through perforated piping network suitably clamped from the bottom. Small tapping with valve to be provided in the steam line near the tank for proper steam flow. Preferably, two number of taps to be provided due to the more possibility of steam choking, so intermediate operation of these valves are required frequently for release of choke.

8.4.6 Preparation of Acid Solution in tank:

For addition of sulphamic acid in the tank, suitable platform with railing is to be provided on the tank. The platform should be capable of holding 15 -20 bags at a time. For safety, ladders with railing and platform along two sides of the tank are also to be provided in such a way so that the addition of sulphamic acid bags can be carried out safely and conveniently. For lifting of acid bags on the platform suitable arrangements are to be made. Sufficient safety measures

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are to be followed while handling Sulphamic acid. Persons should wear goggles, gloves, gumboots, safety belts etc. while working.

For hydrochloric acid, Acid from the acid tanker (approx.-30-33%) can be directly unloaded into the mixing tank in phased manner to maintain the required acid strength in the mixing tank with all safety checks in place. Dedicated Unloading Pumps (1W+1S) should be installed for unloading of Hydrochloric Acid.

Suitable draining arrangements shall be provided to lead the effluents to neutralizing pit to facilitate complete draining of the system.

Alternate arrangement of interrupted power supply (DG set) may be provided for running at least one circulating pump and area lighting for smooth and uninterrupted operation of the process. A Portable Water analysis laboratory with testing instruments as per the testing requirement will be set up near the temporary system.

9.0 Precautions to be taken for Condenser Cleaning:

SAFETY: All necessary safety precautions need to be planned and PPEs need to be available at site (vide safety guidelines mentioned in Annexure-1. Safety showers & MSDS should be available at chemical cleaning area and near to acid storage and handling system.

Before starting chemical cleaning of condenser, the availability of required quantity of chemicals, DM water, clarified water, power supply, steam etc. must be ensured. A hydro test at the design pressure for Chemical Cleaning with DM water should be compulsorily carried out before the commencement of actual cleaning.

The spent acid after chemical cleaning must be neutralized properly before draining off to the N pit. Hydrated lime is used to neutralize the remaining acid during the process.

During chemical cleaning the adherence to the safety regulations for handling acid and other chemicals shall be strictly followed.

A checklist is to be filled up before commencement of Condenser Chemical Cleaning as per Annexure-2.

A mock trial should be conducted prior to the commencement of the condenser cleaning to check the healthiness of the temporary erected system and to acquire an idea of operation of various valves, pumps, and other connected system, so that actual operation is carried out to the nearest accuracy.

9.1 Estimated quantity of acid calculation and Procedure for Chemical Cleaning of condenser:
Before the start of condenser acid cleaning, it is advised to keep a record of CW System Details as per Annexure-5.

Calculation of Estimated quantity acid required for removal of scale deposit in condensers:

For average thickness of scale deposited, boroscopic determination is to be done. Thickness measurement is to be done by site EEMG in presence of site TMD, site Operation and site Chemistry.

Calculation of the approximate acid requirement quantity for Condenser Chemical cleaning is summarized below in Table-III:

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Table-III:		
ID of condenser tube(A)	M	d
Length of condenser tube(B)	M	L
Inner surface area of tube (C)	M2	$2 \times 3.14 \times (d/2) \times L$
Total no. of tubes (D)	Nos	N
Total inner surface area of all tubes(E)	M2	(C)*N
Avg. thickness of scale deposited (F) (RHS/LHS)	M	T
Volume of scale (G)	M3	$T \times (E)$
Density of scale (H)	MT/ M ³	2.8 (for CaCO ₃)
Wt. of scale (I)	MT	$(G) \times (H)$
Wt. of scale in X% of tubes (J)	MT	$(I) \times (X/100)$
Quantity of acid required for Chemical cleaning of condenser@200% (K)	MT	2*J
Acid required (keeping 5% margin) (L)	MT	$(K) \times 1.05$

10.0 INHIBITOR EFFICIENCY TEST:

To be done for acid inhibitors prior to the commencement of Chemical Cleaning:
Initially weighed metallic coupons (of same metallurgy as condenser tubes and Mild Steel) is immersed in inhibited acid for a period of minimum 8 to 10 hours, at a temp of 60 deg C.
Final weighment done after 8 to 10 hours to check the corrosion rates in MPY.
Similarly, determine the corrosion rate in Un-inhibited acid solution of a pre-weighted coupon for the same period.

Determine the efficiency of corrosion inhibitor by the following equation:

$$\text{Inhibitor Efficiency \%} = \frac{\text{Corrosion rate (Without inhibitor)} - \text{Corrosion rate (with Inhibitor)}}{\text{Corrosion rate (without Inhibitor)}} \times 100$$

Inhibitor efficiency should be 99% or more for all metallurgy associated with condenser system.

11.0 General Chemical Cleaning Procedure for Condensers in Power Plant:

It is suggested that the following Off-load method should be adopted during the chemical cleaning of Condenser:

Check the temporary chemical cleaning circuit consisting of circulating pumps, tanks, valves, temporary piping etc. hooked up to the condenser. (General scheme given in Annexure 3).

It is suggested that the cleaning be carried out in stages, in one stage only one pass of the condenser should be cleaned. The acid cleaning for top-to-top, bottom-to-top, and top-to-bottom circulation provision should be carried out as per Annexure-3&4.

Chemical Cleaning may be carried out using Inhibited Sulphamic acid (TM Grade) or Hydrochloric acid (IS-265-Technical Grade) as per the recommended procedure.

11.1 Chemical Cleaning Procedure for Condensers in with inhibited Sulphamic Acid:

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The procedure of chemical cleaning includes the following major steps:

11.1.1 COLD WATER FLUSHING: Fill the system with the DM Water and circulate for 1 hour to ensure no leaks in CW line blanking and temporary piping including the vent line as per the Scheme. Flush the system continuously till the water gets clear at the outlet.

If possible, suspend a portion of the pre-weighed scaled tube in the tank before the next step.

Completion criteria: Circulating water turbidity <10 NTU.

11.1.2 TEMP RAISING AND HOT WATER FLUSHING: Drain the condenser completely. Fill the system with DM water ensuring filling of condenser up to 3/4th of the total volume. Raise the temp to 60 – 65 °C by passing Steam. During heating the water should be circulated using suitable capacity pumps and a temporary Tank of at least 15 - 20 M³ capacity. Check the Turbidity of the sample at the outlet point.

Completion criteria: Circulating water turbidity <10 NTU.

11.1.3 ACID INJECTION:

For Sulphamic acid cleaning, start introducing the Inhibited Sulphamic acid in the system monitoring the acidity level. Generally, the concentration of the acid is to be maintained between 4.0 to 4.5 % w/w. As soon as the desired level is attained, stop the acid dosing. Circulation is to be maintained in the system. During the period, the temperature of the solution is to be maintained between 65 to 70°C by passing steam and circulation is to be continued. The cleaning should be initiated from top side of the condenser. After about 5-6 hours of circulation or when hardness values gets stabilized, interchange the flow direction in other mode of circulation to ensure proper cleaning.

Top-to-top, bottom-to-top and top-to-bottom circulations are to be ensured for effective removal of scales.

Duration of the process will depend upon total hardness, total iron, Acid Strength stabilization on three consecutive readings. Once it is stabilized, circulate the same solution for one hour.

11.1.4 Chemical Parameter Monitoring: Measure pH, Conductivity, Iron & hardness level in the inlet & outlet points during entire cleaning process at an interval of 20-30 minutes (When both inlet & outlet readings are almost equal, values said to be stabilized). Record for the same is to be maintained and submitted along with the Condenser Acid Cleaning Report (joint protocols to be maintained). For test methods, refer Annexure-5. Drain the Solution and Rinse with DM water. During the cleaning process, Vent pipes should be preferably of 4 " diameter both at front and back to release the generated Carbon dioxide in the process.

11.1.5 WATER FLUSHING: Fill the system with fresh water and circulate for ½ an hour and drain the system. Repeat the process of filling, circulation and draining for 3 times to obtain inlet and outlet pH to be same or flush the system continuously till the pH attains neutral.

11.1.6 NEUTRALISATION: Neutralize the system by circulating alkaline solution (approx.-1% Soda Ash solution) to achieve pH of the solution (>7.0). Circulate the solution for 1 hour to remove acid packets in the system. Flush the system. Fill system with fresh water, circulate for ½ hour, and drain the system. Repeat the process of filling, circulation and draining for at least 3 times for obtaining the same inlet and outlet pH. Drain the Solution.

11.1.7 NEUTRALISATION OF SPENT ACID: Acid drained from the system will be neutralized by disposing it into nearby trench or channel along with plenty of water and lime power.

Alternatively, the chemical effluents may be sent to ETP through temporary drain lines erected.

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11.1.8 MECHANICAL (HP JET) CLEANING:

After Chemical cleaning, HP Jetting will be carried out to remove loose scales present in the condenser tubes.

11.1.9 INSPECTION: In order to adjudge the effectiveness of cleaning, Mandrel clearing and/or Boroscopic inspection shall be carried out jointly in presence of outsourced agency, site Chemistry, site TMD, site Operation and site EEMG and joint protocols are to be signed with all details.

In case the above tests are not passed, again jet cleaning is to be repeated.

11.2 Chemical Cleaning Procedure for Condensers in Power Plant with Hydrochloric Acid:

The Chemical Cleaning Procedure involves the following major Steps:

11.2.1 COLD WATER FLUSHING: Fill the system with the DM Water and circulate for 1 hour to ensure no leaks in CW line blanking and temporary piping including the vent line as per the Scheme. Flush the system continuously till the water gets clear at the outlet. If possible, suspend a portion of the pre-weighed scaled tube in the tank before the next step.

Completion criteria: Circulating water turbidity <10 NTU.

11.2.2 TEMP RAISING AND HOT WATER FLUSHING: Drain the condenser completely. Fill the system with DM water ensuring filling of condenser up to 3/4th of the total volume. Raise the temp to 60 – 65 °C by passing Steam. During heating the water should be circulated using suitable capacity pumps and a temporary Tank of at least 15 - 20 M3 capacity. Check the Turbidity of the sample at the outlet point.

Completion criteria: Circulating water turbidity <10 NTU.

11.2.3 ACID INJECTION: Fill the system with fresh water and circulate. Add Corrosion Inhibitor concentration 0.3% of the total volume and circulate for 1 hour. In continuation of the process, to the same solution slowly add Hydrochloric Acid solution and maintain acid strength 2% -3% w/v. Keep the solution under circulation for a period of 5-6 hours (maximum) to loosen the scale to the maximum extent.

After about 5-6 hours of circulation or when hardness values gets stabilized, interchange the flow direction in other mode of circulation to ensure proper cleaning.

Top-to-top, bottom-to-top and top-to-bottom circulations are to be ensured for effective removal of scales.

Duration of the Chemical Cleaning will depend upon the total hardness, total iron, Acid Strength stabilization on three consecutive constant readings. Once it is stabilized, circulate the same solution for one hour.

11.2.4 Chemical Parameter Monitoring: Measure the pH, Conductivity, Iron, and hardness salts level in the inlet and outlet sampling points during the entire cleaning process at an interval of 20-30 minutes (When both inlet & outlet readings are almost equal, values are stabilized). Record for the same is to be maintained and submitted along with the Condenser Acid Cleaning Report (joint protocols are to be maintained). For test methods, refer Annexure-5.

Now drain the Acid solution. During the cleaning process, Vent pipes should be preferably of 4 " diameter both at front and back to release the generated Carbon dioxide in the process. (Refer the Scheme).

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11.2.5 WATER FLUSHING: Fill the system with fresh water and circulate for ½ an hour and drain the system. Repeat the process of filling, circulation and draining for 3 times to obtain inlet and outlet pH to be same or flush the system continuously till the pH attains neutral.

11.2.6 NEUTRALISATION: Neutralize the system by circulating alkaline solution (approx.1% Soda Ash solution) to achieve pH of the solution (>7.0). Circulate the solution for 1 hour to remove acid pockets in the system. Flush the system. Fill system with fresh water, circulate for ½ hour, and drain the system. Repeat the process of filling, circulation and draining for at least 3 times for obtaining the same inlet and outlet pH. Drain the Solution.

11.2.7 NEUTRALISATION OF SPENT ACID: Spent Acid drained from the system will be neutralized by disposing it into the nearby trench or canal along with plenty of water and lime power.

Alternatively, the chemical effluents may be sent to ETP through temporary drain lines erected.

11.2.8 MECHANICAL (HP JET) CLEANING: After Chemical cleaning, HP Jetting will be carried out to remove loose scales present in the condenser tubes.

INSPECTION: In order to adjudge the effectiveness of cleaning, Mandrel clearing and/or Boroscopic inspection shall be carried out jointly in presence of outsourced agency, site Chemistry, site TMD, site Operation and site EEMG and joint protocols are to be signed with all details.

In case the above tests are not passed, again jet cleaning is to be repeated.

NOTE: As condition of the condenser tubes is not known, leakage may develop at some points of the condenser tubes where because of already damaged tubes covered with scale products, may get exposed during acid cleaning. Hence Condenser flood test is to be done post Inhibited Sulphamic acid and hydrochloric acid cleaning as per the approved procedure. Before filling water in the condenser, all tubes are to be dried by air completely. After filling the water to the required level, system shall be kept for minimum 8 hrs before condenser tube inspection for seepage. Any such tubes are to be plugged with the same condenser material.

12.0 RECORDS TO BE MAINTAINED:

S. No.	Record Description	Maintained by	Frequency	Remarks
1	As section 5.2,5.2.1,9.1 annexure-2, annexure-5	Chemistry	As mentioned in the relevant sections	

13.0 DISTRIBUTION OF LMI

A soft copy of the LMI shall be uploaded in the EEMG dept. website on Simhadri intranet LAN by the EEMG dept. The uploaded soft copy of the document is considered as Controlled Copy.

14.0 REVIEW OF LMI

Head of Chemistry is responsible for put up approval from Competent Authority and for revision of LMI on a 2-yearly basis or as and when required.

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CHECKLIST-I
CONDENSER CHEMICAL CLEANING

Sl. No	Description	Compliance (Yes/No)
1	Completion of all overhauls work in pressure parts. Hydraulic test is completed.	Yes/ No
2	The chemical cleaning arrangements viz tanks pipelines, pumps, and pressure gauges, temperature gauges are completed as per the schematics. Hydraulic tests of all temporary pipes, valves and acid circulation pump done.	Yes/ No
3	CW outlet header is made dummy.	Yes/ No
4	Insulation of chemical mixing tank, steam pipe, pump supply and return line completed.	Yes/ No
5	All necessary sampling system like Condenser filling line, return line and at discharge of Condenser outlet) are available as per Scheme	Yes/ No
6	To ensure flushing of each header and system, arrangement to be made for drain line thoroughness checking and flushing (in case of two isolating valves in drain line, line to be cut between two valves otherwise extra valve to be provided).	Yes/ No
7	Inhibitor efficiency test done in laboratory & result found satisfactory.	Yes/ No
8	The expiry date of the inhibitor is to be checked before use.	Yes/ No
9	All chemicals as per the procedure are available at site.	Yes/ No
10	Testing of chemicals as per the requirement is completed.	Yes/ No
11	Neutralization and effluent disposal arrangements are available.	Yes/ No
12	Safety arrangement:	Yes/ No
	a. Safety shower and eye wash system near acid handling area.	Yes/ No
	b. Fire water and sufficient fire hoses at chemical handling area	Yes/ No
	c. medical facilities/ ambulance and first aid box at site for chemical burns.	Yes/ No
	d. Firefighting personnel's during the cleaning process.	Yes/ No
	e. Safety appliances for personnel at site.	Yes/ No
	f. Sufficient water availability at chemical handling area for flushing, washing etc.	Yes/ No
	g. A mock drill to be conducted under the supervision of safety for the preparedness of above points before the start of acid dosing activity.	Yes/ No
	h. Proper illumination of chemical handling area, N-pit and Condenser area.	Yes/ No
	i. Communications arrangements at chemical handling, Condenser area, and effluent disposal areas.	Yes/ No
	j. Telephones and/or walkie- talkie at Chemical Cleaning location	Yes/ No

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	k. Steam and DM water lines for solution preparation cum circulation tank are completed.	Yes/ No
	l. Flushing water for acid pumps availability at site.	Yes/ No
	m. Chemical cleaning activity board and logbooks availability.	Yes/ No
	n. Availability of MSDS of all the chemicals at use at the locations	Yes/ No
	o. Adequate DM water availability for chemical cleaning activity.	Yes/ No
	p. Emergency power supply for one circulation pump from DG set.	Yes/ No
13	Mock drill for following emergency conditions and action plan:	Yes/ No
	a. Chemical spillage	Yes/ No
	b. Chemical leakages	Yes/ No
	c. Undesired chemical reaction during process.	Yes/ No
	d. Failure of valves, pumps, and pipelines	Yes/ No
14	Chemical mixing tanks with steam heating coil, acid storage tanks and transfer pumps of adequate capacity are available.	Yes/ No
15	Acid unloading arrangement from tankers (use only pumps for unloading) for Hydrochloric acid	Yes/ No
16	Acid and other dosing pumps are available.	Yes/ No
17	Arrangement of temporary laboratory with white light facility near to chemical cleaning area.	Yes/ No
18	Testing facilities near cleaning site should be developed for testing of parameters like acidity, Fe, Total Hardness, Calcium Hardness etc as per the procedures.	Yes/ No
19	Isolation of all instruments and sampling lines in Condenser as applicable	Yes/ No
20	Formation of chemical cleaning team consisting of Operation, EEMG, Maintenance, & Chemistry.	Yes/ No

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CHECKLIST -II

Sl No	Item description	Reference clause/section in OGN-OPS-CHEM-009	Responsibility*	Compliance (Yes/No)
1.	Cooling Water System parameter monitoring	Table-III, section 4.4	Chemistry, EEMG	Yes/No
2.	Online monitoring of pH, Specific conductivity and turbidity and hooking to PI server	Section 4.4	C&I, Chemistry	Yes/No
3.	Online monitoring of deposition, corrosion rate and biofouling; and record keeping	Table-IV, Section 4.5	Chemistry	Yes/No
4.	Microbiological analysis	Table-V & VI, Section 4.6	Chemistry	Yes/No
5.	Monthly Jar Test of Raw Water	Section 4.7	Chemistry	Yes/No
6.	COLTCS operation	Section 4.7	Operation, TMD	Yes/No
7.	CW system preservation during RSD	Section 4.8	Operation, Chemistry, MMD-OFS, EEMG	Yes/No
8.	Chemical Cleaning of condenser tubes (including Annexure-2- compliance, condenser scale thickness measurement)	Section 5.0 & Annexure-2	Operation, TMD, Chemistry, EEMG, MMD-OFS	Yes/No
9.	Condenser Flood Test	Section 5.7.5	Operation, TMD, Chemistry, EEMG	Yes/No

Signature:

Chemistry

TMD

EEMG

MMD-OFS

Operation executive

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ANNEXURE-1: SAFETY GUIDELINES

Hazards:

High temperature fluid and steam

High concentration chemicals including hazardous acids.

High temperature acid solutions

Electrical hazard

Slippery surface

Working at heights

Frequent movement along the staircase

Chemical handling:

Acid leak from pumps, couplings, pipelines

Acid/chemical leak as jet from pinholes

Fire hazard.

Mitigation measures:

Good quality leak proof pumps and connections.

Availability of PPEs like, full-body apron, laboratory aprons, safety goggles (with full eye cover), safety shoes, hard helmet, acid resistant hand gloves, safety shower & eye wash at the location.

Good lighting.

Good drainage of the area.

Availability of MSDS of all the chemicals at use at the locations.

First aid, availability fire hydrant

Valve tagging and direction of flow, with full visibility of open or close condition.

Pep talks and display of other information at the site.

Close supervision and individual training for the workers.

Full time availability of hospital emergencies during the process.

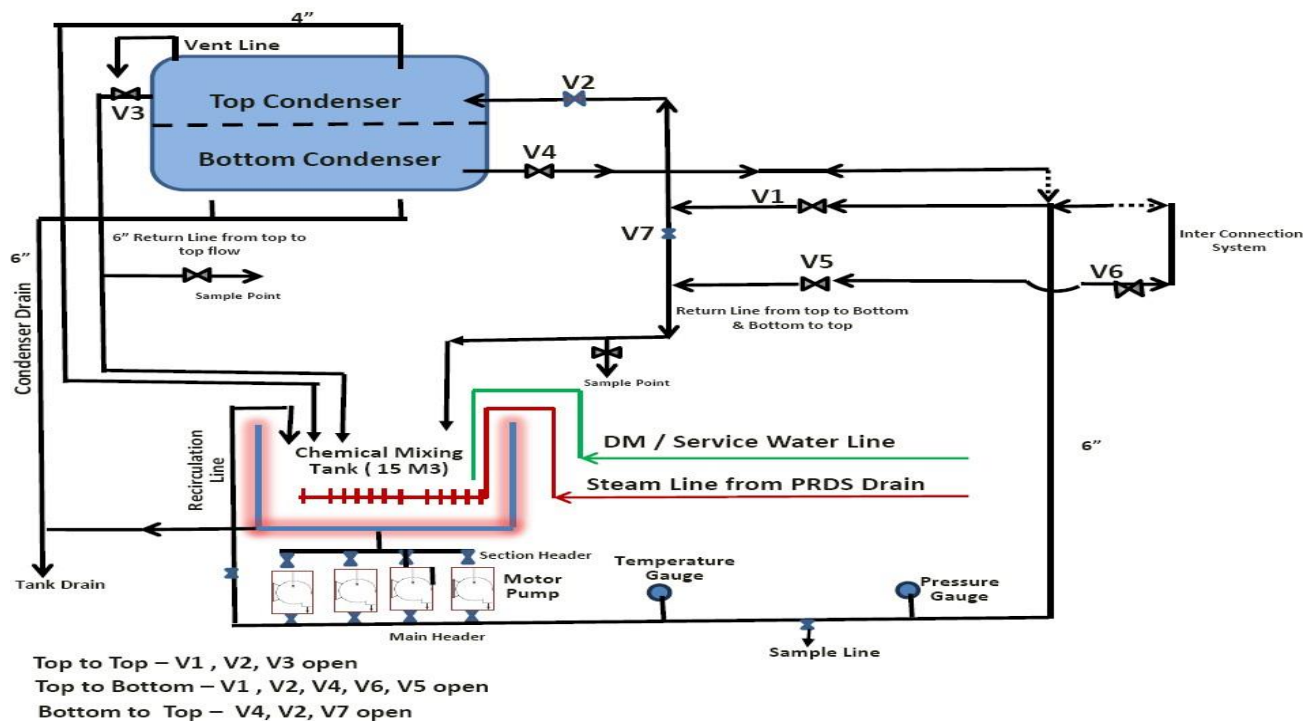
Contractors should submit documents regarding the emergency handling, procedure covering the aspects like Open Flames, Electric Welding, Spark Restrictions, and Fire Equipment, Leaking Piping, Valves, & Fittings, Leak Repair During Cleanings, Neutralization Containment of Chemical Spills, Emergency Pump Shutdown Procedures in its bidding document.

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ANNEXURE-2:

ILLUSTRATIVE LINE DIAGRAM FOR CONDENSER ACID CLEANING IN 03 CIRCULATION MODES- SUB-CRITICAL UNITS

Single Line Diagram: Sub-Critical Units



ANNEXURE-3

CONDENSER/CW TREATMENT SPECIFICATIONS

Station-

Unit-

Sl No	Parameters	Values	UOM
1.	Design C.W. Temperature		0C
2.	C.W. Temperature rise		
3.	Condenser Back pressure		mm Hg
4.	C.W. flow quantity		m ³ /hr
5.	C.W. side pressure drop		mwc
6.	No. of C.W. passes		
7.	No. of tubes		
8.	Tube dimensions:		
	a) Tube ID		mm
	b) Length of tubes		mm

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9.	Tube material:		
	a) Condensing zone		
	b) A/C zone		
10.	Water box design pressure		
11.	Water box Hydraulic test pressure		kg/cm ² (g).
12.	Others		

ANNEXURE-4:

ANALYSIS METHODS FOR DETERMINATION OF CHEMICAL PARAMETERS DURING ACID CLEANING:

DETERMINATION OF ACID CONCENTRATION IN THE SOLUTION

a) REAGENTS:

Sodium hydroxide solution, standard 1.0 N

Mixed Indicator: 3 parts of Bromo cresol green- 0.1 % solution in alcohol 1 part of methyl red – 0.2 % solution in alcohol

Methyl Orange Solution (0.1%)

b) PROCEDURE FOR DETERMINATION OF SULPHAMIC ACID (%)

Pour 10 ml of test sample into a conical flask

Add two or three drops of mixed indicator

Titrate with 1.0 N NaOH solution. End point from red through gray to blue-green.

c) CALCULATION

$$\% \text{ Acid} = V \times 0.97$$

Where V = milliliters of 1.0 NaOH used to obtain end point.

PROCEDURE FOR DETERMINATION OF HYDROCHLORIC ACID (%)

a) Take 5 ml. sample and dilute with DM water.

Add 3-4 drops of Methyl Orange Indicator. Titrate with Std.1 (N) NaOH Sol.

b) CALCULATION

$$\text{Acid (\%)} = \frac{Y \times N \times 36.5}{5 \times 10} = 0.73 \times Y \quad (Y = \text{ml. of NaOH. (N = Normality of NaOH.)})$$

METHOD OF ESTIMATION OF IRON IN THE ACID SOLVENT SYSTEM USED FOR CHEMICAL CLEANING OF CONDENSER POTASSIUM DICHROMATE METHOD

a) Reagents: Dil. HCl (1:1v/v). Stannous Chloride solution. Mercuric chloride solution (Saturated), Dilute Pot. Permanganate (0.1N), Sulphuric & Phosphoric Acid Mixture.: (150 ml Conc Sulphuric Acid + 700 ml of water) +150 ml of 85% Phosphoric Acid, destroy oxidizable impurities by adding pot. Permanganate sol 0.1N until pink colour of permanganate persists. Sodium diphenyl amine sulphonate indicator sol: Dissolve 0.2 - 0.3gm of Sodium diphenyl amine sulphonate in 100 ml of hot water cool & store in amber colour bottle. Std. Potassium Dichromate Sol.: 0.1N.

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b) PROCEDURE:

Take 50 ml of solution in 500/250 ml conical flask. Add 10 ml of dil HCL.

Add 5 ml of Conc HCl (If required).

Reduce the iron by adding Stannous Chloride sol. in drop wise until the solution becomes colorless, then add 2-3 drops excess.

Add 10 ml of Mercuric Chloride sol. at a time & stir thoroughly.

A small silky white ppt of mercurous chloride should form slowly, indicating that a slight excess of stannous chloride present. Allow to stand 2 – 3 min. until more ppt appears to form.

Add 15 ml of Z-R (Sulphuric + Phosphoric Acid Mix.) reagent & dilute to 100ml.

Add 5 ml of sds indicator sol. & Titrate with Std. Potassium Dichromate sol. with stirring continuously until the GREEN colour begins to darken, continue titration until a drop of Potassium Dichromate sol produces a permanent VIOLET colour.

Determine indicator blank.

- c) Calculation: Iron (as Fe) % by weight: Vol. of std Pot. Dichromate (0.1N) for test - Vol. of std Pot. Dichromate (0.1N) for Indicator blank * 5.584 / wt in gm of original solution.

METHOD OF DETERMINATION OF HARDNESS IN THE ACID SOLVENT SYSTEM USED FOR CHEMICAL CLEANING OF CONDENSER

a) REAGENTS

Ammonium hydroxide solution (1+4)

Mix 1 volume of NH₄OH (sp. gr. 0.90) with four volumes of water.

Buffer solution – Prepare the buffer solution in three steps as follows: Dissolve 40 g of sodium tetra borate in 800 ml of water.

Dissolve 10 g of sodium hydroxide (NaOH), 10 g of sodium sulphide (Na₂S.9H₂O), and 10 g of potassium sodium tartrate (KNaC₄O₆.4 H₂O) in 100 ml of water.

When cool mix the two solutions and add 1 g of magnesium disodium ethylenediamine tetra acetate. Make up to 1L with water.

Standard Calcium solution 91 ml = 0.20 mg CaCO₃)

Dissolve 0.2000 g of CaCO₃ in 3 to 5 ml of HCl (1+4). Dilute to 1 litre with water.

Calcium Indicator

Use powdered hydroxy naphthol blue/grind solid hydroxy naphthol blue to 40 to 50 mesh size.

Hardness Indicator Powder

Grind 0.5 g of Chrome Black T with 100 g of powdered sodium chloride. Use a dark- coloured bottle for storage.

Standard EDTA solution (1 ml = 1.0 mg CaCO₃)

Dissolve 3.8 g of disodium ethylenediamine Tetraacetate dehydrate (Na₂H₂ EDTA) in approx. 800 ml of water. Adjust the pH of the solution to 10.5 with NaOH solution (50 g/l). Determine the concentration of this solution using the standard calcium solution. Adjust the concentration of the EDTA solution so that 1 ml will be equivalent to 1.0 mg of CaCO₃. Store the standard EDTA solution in polyethylene bottle.

Sodium hydroxide solution (50 g/l)

Dissolve 50 g of sodium hydroxide in water and dilute to 1 litre.

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b) PROCEDURE

Hardness – Measure 50 ml of clear sample into a clear colorless conical flask. Adjust the pH of the sample to 10 by adding Ammonium hydroxide solution. Add 0.5 ml of buffer solution and approx. 0.2 g of hardness indicator powder. Titrate with standard EDTA solution until the colour changes from red to blue. Calcium Hardness - Measure 50 ml of clear sample into a clear colorless conical flask. Add 2 ml NaOH solution and stir. Add approx. 0.2 calcium indicators and stir. Titrate with standard EDTA solution until the colour changes from red to royal blue.

c) Calculations:

Hardness, mg/l as = $20 \times V$ (V = Volume of Standard EDTA solution added)

LIST OF CHEMICALS FOR ANALYSIS

- 1.pH Paper
- 2.Standard solution ampules
- 3.Sodium Hydroxide (1.0 N)
- 4.Sodium thiosulphate (0.1 N)
- 5.EDTA (0.1 N)
- 6.Indicators
 - Bromo cresol green
 - Methyl red
 - Starch
 - Phenolphthalein
 - Methyl orange
 - Chrome black T
 - Hydroxy-naphthol blue
- 7.Ammonium Bifluoride
- 8.Hydrazine Hydrate
- 9.Potassium Iodide
- 10.Potassium Permanganate
- 11.Sodium Hydroxide
- 12.Sodium Tetra Borate
- 13.Sodium thiosulphate
- 14.Ammonium hydroxide
- 15.Sodium Chloride
- 16.Potassium Sodium tartrate
- 17.Magnesium Disodium-ethylenediamine tetra acetate